

3.2.6 Lesson Review

Date: 3/13/2026, 8:47:41 PM

Time Spent: 08:06

Score: 100%

Passing Score: 80%



Question 1

✓ Correct

Which of the following are examples of data compliance and regulatory programs mentioned as important for AI data security controls? (Select three.)



Payment Card Industry Data Security Standard (PCI-DSS)



Correct



OpenAI Model Licensing Agreement



Health Insurance Portability and Privacy Act (HIPAA)



Correct



General Data Protection Regulation (GDPR)



Correct



Fair Use Doctrine



Data Masking Protocol (DMP)



National Security Agency Encryption Mandate

Explanation

General Data Protection Regulation (GDPR) is a correct answer. GDPR is a key regulatory compliance program from the European Union important for protecting and regulating data security for AI systems.

Health Insurance Portability and Privacy Act (HIPAA) is a correct answer. HIPAA is a United States regulation that governs how medical information is protected and is a regulatory program relevant to AI data security.

Payment Card Industry Data Security Standard (PCI-DSS) is a correct answer. PCI-DSS is an industry standard that sets requirements for organizations handling credit card information; it is also a compliance program that impacts AI data security controls.

OpenAI Model Licensing Agreement is incorrect. This is not a regulatory compliance program for data protection; it is a hypothetical licensing agreement unrelated to industry-recognized data protection standards.

Fair Use Doctrine is incorrect. Fair Use is a legal doctrine related to copyright law and usage rather than a data compliance or regulatory program governing data security for AI.

Data Masking Protocol (DMP) is incorrect. While data masking is a protective technique, Data Masking Protocol is not a recognized regulatory compliance program.

The National Security Agency Encryption Mandate is incorrect. This mandate is not a recognized regulatory compliance or data protection program in the industry. Regulatory programs typically encompass industry-wide standards, rather than agency-specific directives.

Related Content

resources\questions\q_overview_of_ai_data_security_controls_02.question.xml

Question 2

✓ Correct

Which of the following technologies is specifically designed to help protect data while it is being actively processed or used by an AI system?

- ☐ Secure Sockets Layer (SSL)
- ☐ Data classification labeling (such as "Confidential" or "Top Secret")
- ☒ Trusted Execution Environment (TEE) ✓ Correct
- ☐ File-level encryption

Explanation

Trusted Execution Environment (TEE) is correct. A TEE is a secure processor area that protects data while it is actively being processed ("in use"), keeping sensitive data isolated from the rest of the system.

File-level encryption is incorrect. It secures data at rest but does not protect it during processing.

Secure Sockets Layer (SSL) is incorrect. SSL encrypts data in transit, not during CPU or AI system processing.

Data classification labeling (such as "Confidential" or "Top Secret") is incorrect. Labels indicate data sensitivity and handling requirements, but do not themselves protect or encrypt data in use. Actual protection during processing requires approaches like TEE.

Related Content



3.2.2 Encryption of AI Data

resources\questions\q_encryption_of_data_05.question.xml

A financial services company is building an AI system to recommend credit products to its customers.

During development, the team notices that the AI's recommendations are consistently favoring certain customer groups over others, resulting in unequal credit product offers. You are asked to apply data safety measures to address the risks linked to bias and the quality of data input.

Which of the following actions should you take FIRST to address this issue and improve the fairness of the AI model?

- ☒ Review and clean the training data set to identify and remove biased or poor-quality data before retraining the AI system. ✓ Correct
- ☐ Encrypt all emails between the data science team and IT operations to protect discussions about the model's architecture and customer data.
- ☐ Reiterate to the AI development team that malicious data leaks must be reported promptly according to the company's data breach policy.
- ☐ Strengthen encryption on all stored AI models so that unauthorized access to the recommendation outputs is prevented.

Explanation

Reviewing and cleaning the training data to remove biased or poor-quality data before retraining the AI system is correct. This directly improves input data quality and reduces bias, ensuring AI models generate fair and accurate results. High-quality, representative data is essential for mitigating unintended bias.

Reminding staff to promptly report malicious data leaks is incorrect. While reporting breaches is important, this does not address the fairness or quality of training data, but focuses on incident response.

Strengthening encryption on stored AI models is incorrect. Encryption protects against unauthorized access, not against bias or poor-quality training data, and does not improve model fairness or accuracy.

Encrypting team emails is incorrect. While this secures communication, it does not address data quality or bias in AI training data.

Related Content

resources\questions\q_data_safety_measures_03.question.xml

Question 4

✓ Correct

What is the primary benefit of encrypting data in transit in AI systems?

- ☒ It prevents unauthorized interception and reading of sensitive data sent over a network. ✓ Correct
- ☐ It anonymizes all user data before any processing occurs.
- ☐ It increases the computational speed of the AI system when handling user requests.
- ☐ It protects data from being modified while stored on a physical hard drive.

Explanation

It prevents unauthorized interception and reading of sensitive data sent over a network is the correct answer. Encrypting data in transit ensures that sensitive information traveling between endpoints (such as between an AI system and a user) cannot be intercepted and read by unauthorized parties.

It protects data from being modified while stored on a physical hard drive is incorrect. This describes the protection of data at rest, not data in transit.

It anonymizes all user data before any processing occurs is incorrect. Encryption in transit does not anonymize user data; it only protects its confidentiality while moving between systems.

It increases the computational speed of the AI system when handling user requests is incorrect. Encryption generally adds some computational overhead rather than increasing computational speed.

Related Content



3.2.2 Encryption of AI Data

resources\questions\q_encryption_of_data_03.question.xml

A financial technology company is developing a machine learning (ML) model to detect fraudulent transactions using customer data that includes names, account numbers, transaction amounts, and locations.

To comply with data privacy regulations, the company decides to implement data safety measures before sending any data to the development team for model training.

Which of the following is the MOST appropriate way to apply data masking in this scenario?

- ☒ Replace customer names and account numbers with fictitious, yet consistently mapped, values in the dataset provided to the team while preserving real transaction amounts and locations. ✓ Correct
- ☐ Provide real, unmasked data to the development team but require them to sign a non-disclosure agreement (NDA) to protect customer privacy.
- ☐ Permanently delete customer names and account numbers from all transaction records before sharing the data set with the team, so only transaction amounts and locations remain.
- ☐ Encrypt the entire dataset, including names, account numbers, amounts, and locations, and give the development team the encryption keys to decrypt and use all information.

Explanation

Replacing customer names and account numbers with fictitious, yet consistently mapped, values in the dataset provided to the team while preserving real transaction amounts and locations is correct. Data masking replaces sensitive identifiers with fictional yet consistent values, reducing privacy risk while still letting the team analyze patterns for model training.

Permanently deleting customer names and account numbers from all transaction records before sharing the data set with the team, so only transaction amounts and locations remain, is incorrect. This is data minimization, not masking. Removing identifiers can limit analysis if customer patterns matter for fraud detection.

Encrypting the entire dataset and giving the team the decryption keys is incorrect. Encryption protects data at rest or in transit, not by masking sensitive information within workflows.

Providing real, unmasked data with a non-disclosure agreement (NDA) is incorrect. An NDA is a legal, not technical, safeguard and doesn't prevent data misuse.

Related Content



3.2.3 Data Safety Measures

resources\questions\q_data_safety_measures_04.question.xml

An international healthcare company is developing an AI-powered system to process patient data for improving treatment recommendations. The company operates in regions governed by various data compliance regulations, including the European Union's GDPR and the US HIPAA.

During the design phase, the project manager must choose the most appropriate data safety measure to ensure compliance with these regulations when handling sensitive personal data.

Which action BEST demonstrates compliance with regulatory data safety requirements during the design of this AI system?

- ☒ Classify all patient data using appropriate labels (e.g., confidential, restricted), restrict access based on role, and ensure monitoring and auditing of data handling according to applicable data compliance standards. ✔ Correct
- ☐ Store the entire data set, including identifying details, outside the company's secured network for easier cross-team collaboration in different regions.
- ☐ Allow all development staff unrestricted access to patient data for maximum system functionality, as long as they sign a company's privacy policy.
- ☐ Anonymize all patient data after system deployment, and only apply access controls if a data breach is detected.

Explanation

Classifying all patient data with appropriate labels (e.g., confidential, restricted), enforcing role-based access control, and monitoring data handling per regulations is correct. This follows best practices like GDPR and HIPAA: assigning labels, restricting access, and monitoring access ensures both compliance and patient privacy.

Allowing all development staff unrestricted access to patient data with just a signed privacy policy is incorrect. Legal agreements alone do not fulfill compliance; regulations require technical and procedural safeguards, including limited access and ongoing monitoring.

Storing the entire dataset, including identifiers, outside the secured company network for easier collaboration is incorrect. This violates compliance standards and exposes data to unauthorized access, breaches, and legal risks.

Anonymizing patient data only after deployment and adding access controls only after a breach is incorrect. Access controls and anonymization must be proactive for true compliance.

Related Content



3.2.3 Data Safety Measures

resources\questions\q_data_safety_measures_05.question.xml

A healthcare company is developing an AI-driven application that collects sensitive patient information via web forms and transmits this data to a cloud-based AI model for analysis.

The security team is responsible for ensuring the confidentiality of patient data while it is moving from the user's browser to the cloud infrastructure.

Which of the following is the BEST solution for this requirement?

- ☐ Rely on password protection for user accounts in the web portal, and use HTTP for data transmission.
- ☐ Perform data anonymization before storing the information in the database, and allow data to be sent over plain FTP.
- ☐ Use data masking in the database to obscure sensitive data and require regular password updates for users.
- ☒ Implement TLS 1.3 encryption for all data transmissions between the web application and the cloud AI service. ✓ Correct

Explanation

Implement TLS 1.3 encryption for all data transmissions between the web application and the cloud AI service is correct. TLS 1.3 provides strong end-to-end encryption for data in transit, preventing sensitive information from being intercepted or read during transmission.

Relying on password protection with HTTP for data transmission is incorrect. Passwords restrict access, but HTTP does not encrypt transmitted data, making it vulnerable to interception.

Using data masking in the database and requiring password updates is incorrect. Data masking protects data at rest, not during transmission. Password updates don't secure data as it travels over the network.

Performing data anonymization before storage and sending data over plain FTP is incorrect. Anonymization helps privacy but doesn't guarantee full protection, and FTP leaves data exposed to interception. Encryption in transit is essential.

Related Content

 3.2.2 Encryption of AI Data

resources\questions\q_encryption_of_data_04.question.xml

A healthcare company is developing an AI model to assist medical staff in diagnosing patients using sensitive health records.

During a recent data breach simulation, the company discovered that some patient records were accessed by unauthorized personnel because raw data sets with unredacted personal and medical information were used for AI training.

As the new data security lead, what is the BEST step to apply data safety measures and reduce the risk of data breaches involving sensitive patient records during future AI training?

- ☐ Limit patient record access by physically securing the server room with advanced locks and 24/7 surveillance.
- ☒ Implement data anonymization and data minimization practices before using patient records in AI model training. ✓ Correct
- ☐ Regularly review and update the data privacy policy, making sure all employees electronically sign it to acknowledge receipt and understanding of the policy.
- ☐ Send a company-wide memo instructing employees to never share patient data outside the work environment.

Explanation

Implementing data anonymization and data minimization before using patient records in AI model training is correct. Anonymization removes identifying details, protecting privacy even if data is accessed improperly. Minimization restricts data use to only what's necessary, reducing exposure and breach impact.

Sending a memo instructing employees not to share patient data is incorrect. While awareness is useful, memos alone don't actively control or protect data during AI training.

Physically securing server rooms is incorrect. Physical security protects hardware, but doesn't address digital risks or the use of real, unmasked data in AI models. Sensitive data can still be misused or exposed during processing.

Updating privacy policies for employee acknowledgment is incorrect. Policies alone do not prevent breaches—technical controls are required.

Related Content



3.2.3 Data Safety Measures

resources\questions\q_data_safety_measures_02.question.xml

Your organization is developing a cloud-based AI application that processes and stores sensitive customer information, including personal identifiers.

As a security measure, you are considering how to protect this data while it is stored on the cloud server but not being actively used or transmitted.

Which of the following solutions should you implement to BEST address this requirement?

- ☐ Implement HTTPS to secure communications between users and the AI application, and require strong passwords for system login.
- ☒ Use a file-level encryption tool and store encryption keys in a cloud Key Management Service (KMS). ✓ Correct
- ☐ Apply data masking techniques when the data is requested by users, and rely on operating system access controls for storage protection.
- ☐ Enable multi-factor authentication (MFA) for all user accounts and regularly monitor access logs to detect suspicious activity, in addition to training staff on proper data handling procedures.

Explanation

Using a file-level encryption tool and storing encryption keys in a cloud Key Management Service (KMS) is correct. This encrypts stored data and secures keys, ensuring that data remains unreadable without proper authorization, even if storage is accessed.

Implementing HTTPS and requiring strong passwords is incorrect. HTTPS protects data in transit, not at rest, and strong passwords secure logins but do not encrypt stored files.

Applying data masking on user requests and relying on OS access controls is incorrect. Data masking is for display or processing, not storage. OS access controls alone are insufficient protection if the system is compromised; encryption is stronger.

Enabling multi-factor authentication, monitoring logs, and training staff is incorrect. These improve access security and awareness but do not encrypt data at rest.

Related Content



3.2.2 Encryption of AI Data

resources\questions\q_encryption_of_data_02.question.xml

Question 10

✓ Correct

Which of the following are considered special protection mechanisms used as data security controls for AI systems to ensure regulatory compliance? (Select two.)

- ☐ Ignoring data classification
- ☐ Increasing data storage capacity
- ☒ Encryption ✓ Correct
- ☒ Data masking ✓ Correct
- ☐ Disabling firewalls

Explanation

Encryption is a correct answer. Encryption is a special protection mechanism for AI data. It ensures that information is unreadable to unauthorized users, which is essential for compliance with various regulatory and security standards.

Data masking is a correct answer. Data masking is another special protection mechanism. It involves replacing real data with fictional or obfuscated data to prevent exposure of sensitive information, helping organizations meet compliance requirements.

Increasing data storage capacity is incorrect. While increasing storage capacity may be necessary for handling large datasets, it is not a special protection mechanism for ensuring the security or compliance of AI data.

Disabling firewalls is incorrect. Disabling firewalls actually decreases the security of information systems and exposes them to unauthorized access or attacks; it is the opposite of a protective measure for AI data.

Ignoring data classification is incorrect. Ignoring data classification undermines data security and regulatory compliance efforts, as proper classification is necessary for applying appropriate controls and protection measures to sensitive information.

Related Content

resources\questions\q_overview_of_ai_data_security_controls_01.question.xml